Non-commercial catch estimation for 2023 Deep-7 bottomfish stock assessment

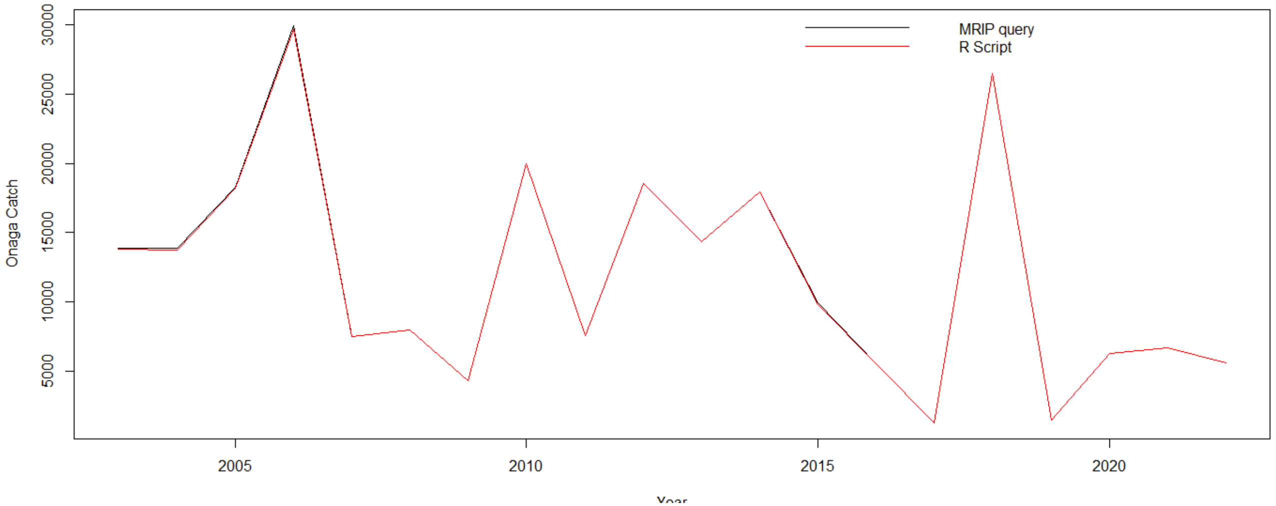
MHI Deep-7 stock assessment working group

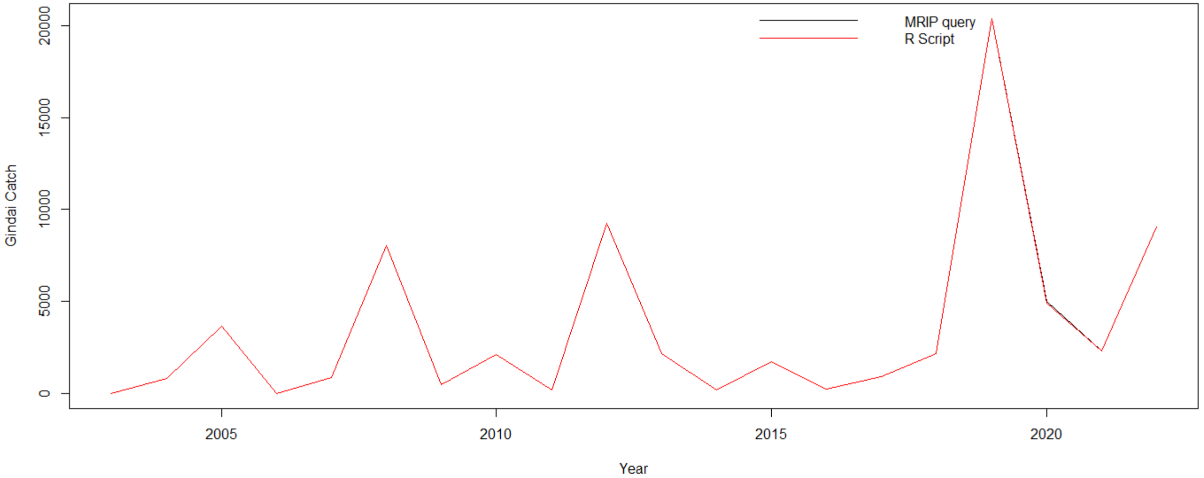
Comparisons of expansion results from R script

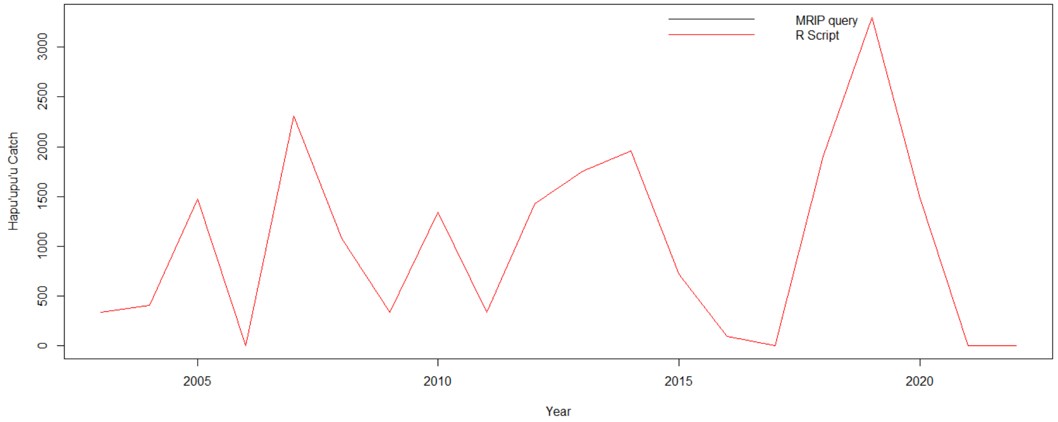
- Total catch (2003-2021) from R expansion vs total catch number from MRIP query (including standard error)

- Proportion of sold catch from the R script vs the proportion of sold catch calculated by Ma et al. (2004-2016)

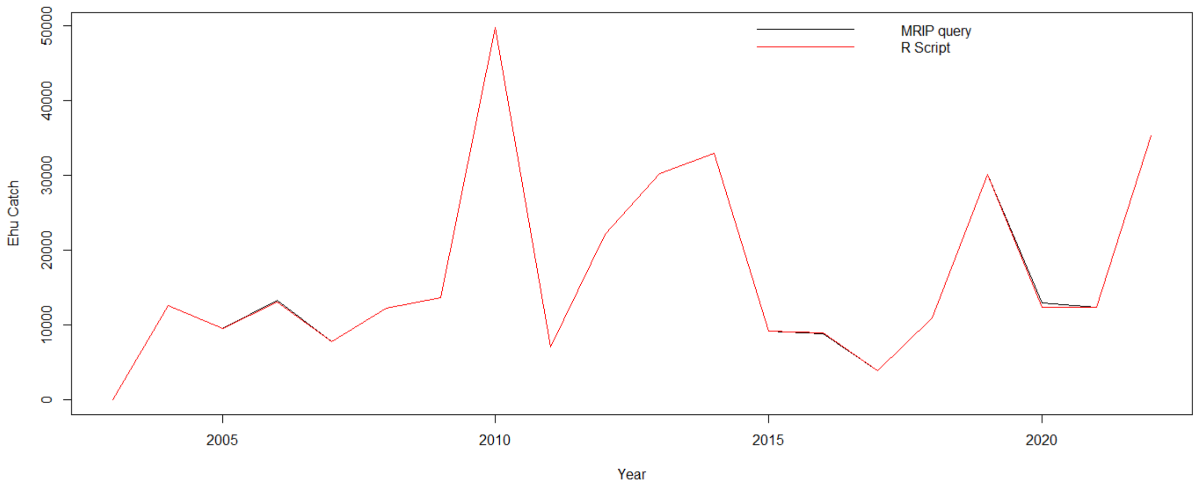
Catch # estimates (2003-2022) from R script (red) vs MRIP query (black): no visible differences for 4 species

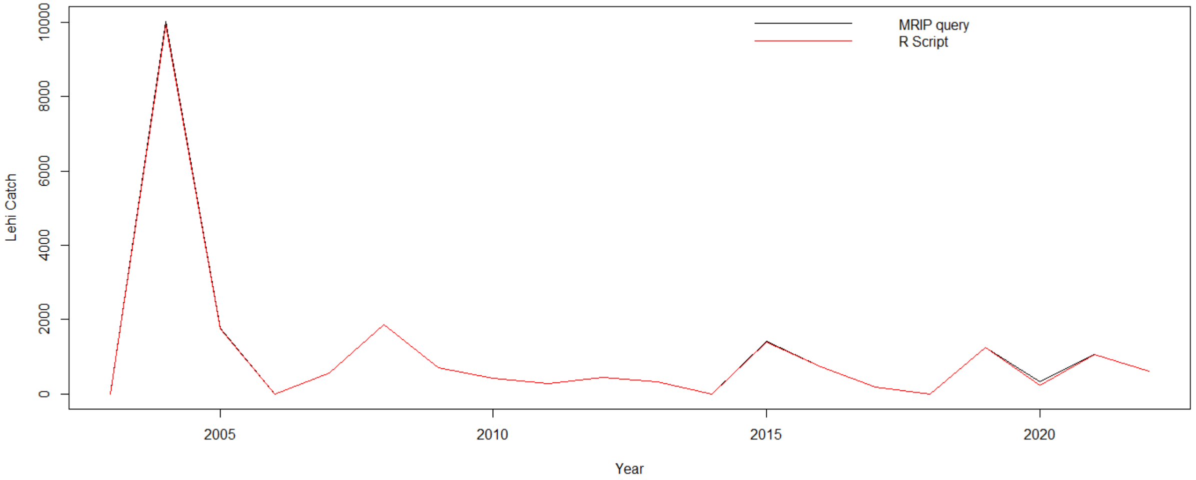




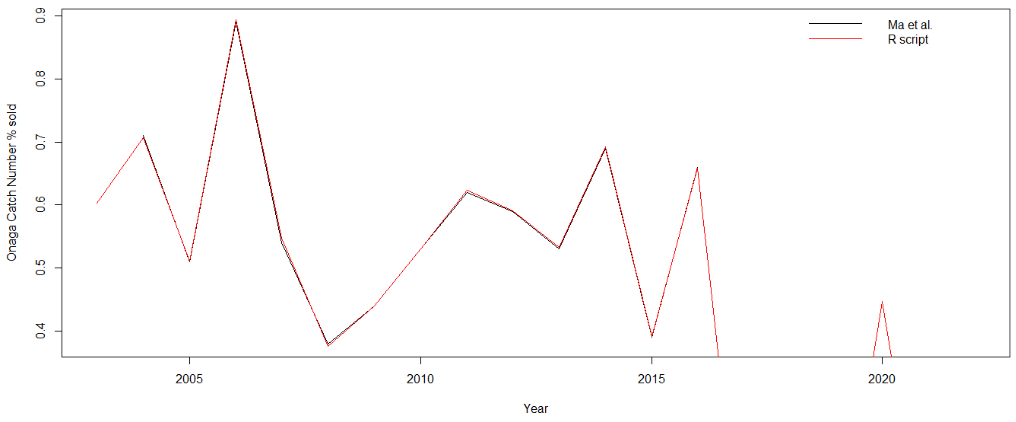


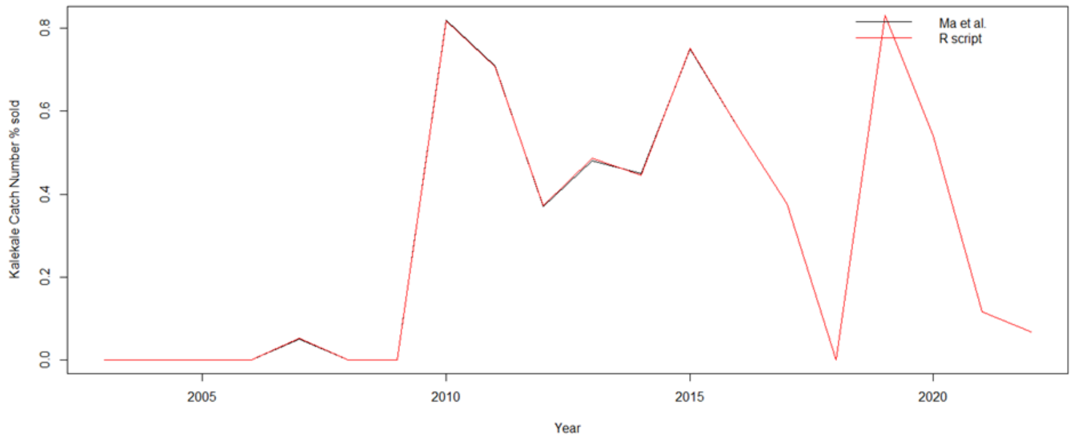
Catch number estimates (2003-2022) from R script (red) vs from MRIP query (black): minor differences in 2020

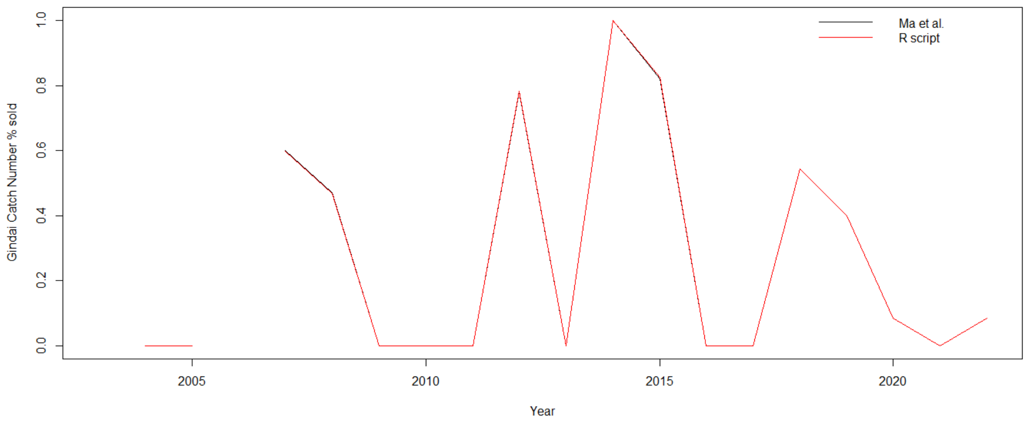


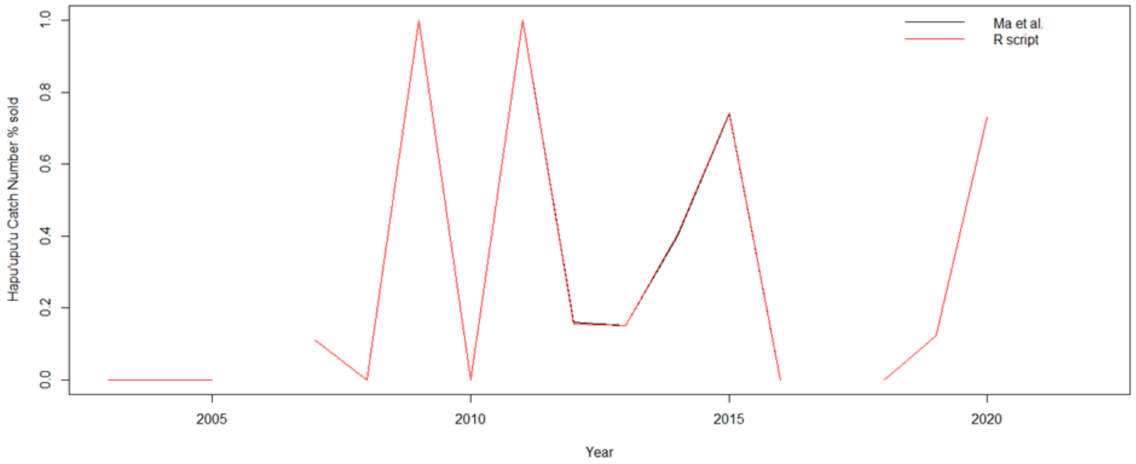


% sold from R (2003-2022, red) vs % sold from Ma et al. (black, 2004-2016): no visible differences for 5 species

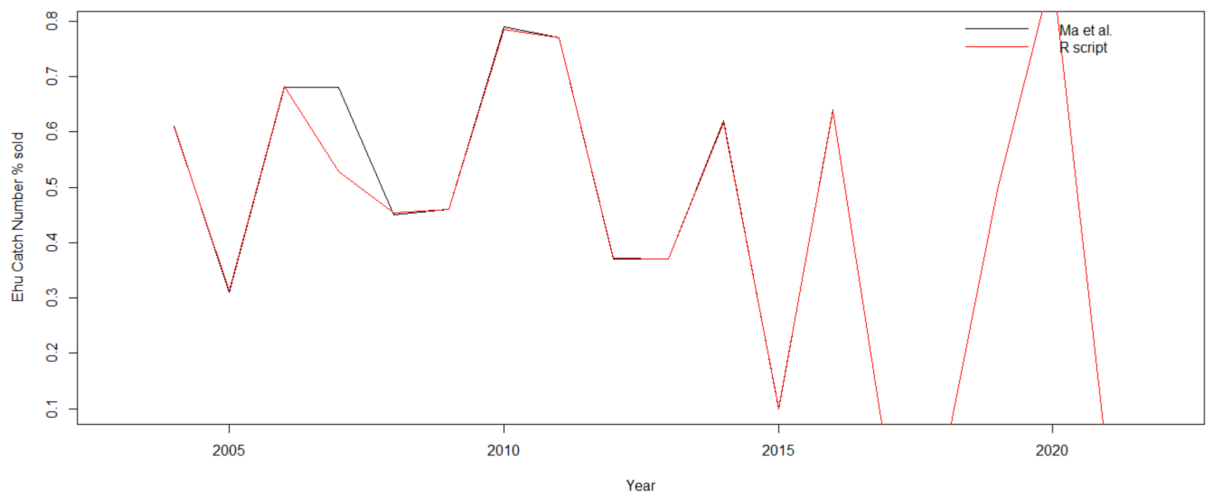


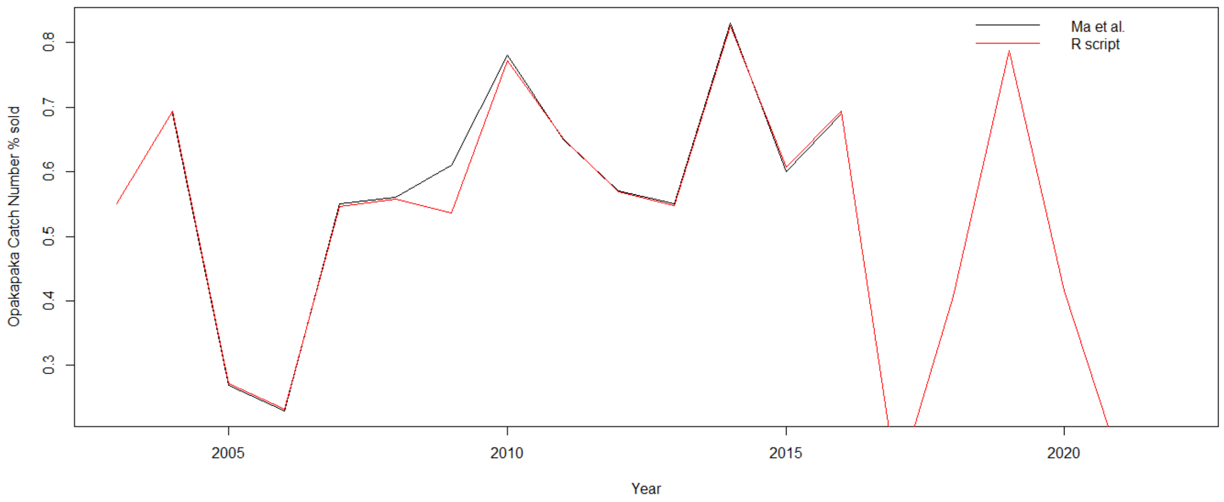






% sold from R vs % sold from Ma et al. (2004-2016): difference in 2009 for paka and in 2007 for ehu





[,:
[,1] [,2] [,3] [,4] [,5] [,6] [,7] [1,] 0.9916861 0.9995629 NaN 0.9982184 0.9940708 NaN NaN [2,] 0.9965580 0.9920630 1.0001969 1.0085315 0.9968651 0.9903956 0.9953322 [3,] 0.99

Ratio of standard errors (SE) from R script vs from MRIP query

	V1	V2	V3	V4	V5	V6	V7
1	0.991427	0.999563		0.998076	0.994071		
2	0.996468	0.992064	1.00014	1.008533	0.996871	0.990396	0.995242
3	0.993307	1.010494	0.993373	1.036887	0.993371	0.987714	0.993894
4		0.989472	0.985265	0.994251	0.995199		
5	1.004663	0.999875	1.009716	1.025902	1.025937	0.998206	0.999675
6	0.999721	0.999764	1.000035	1.022627	0.999961	0.999876	0.999972
7	1.001601	0.999742	1.039059	1.027786	1.02031	0.9994	1.028612
8	0.999967	1.007201	1.000724	1.030777	1.06791	1.001199	1.054243
9	0.998582	1.012991	1.038279	1.023936	1.00723	1.001262	0.998582
10	0.999687	1.003276	1.010746	1.025115	1.010683	0.999244	1.002382
11	1.02895	1.031196	1.066569	1.025828	1.031178	0.998445	1.000525
12	1.011577	1.024385	1.065751	1.014194	0.999567		0.999532
13	0.999661	1.045381	1.038835	1.00228	1.005492	1.021476	0.99861
14	1.001798	1.023552	1.014498	1.00653	1.01704	1.056272	0.999773
15		0.999098	1.021322	1.000086	0.999929	1.001385	0.999453
16	1.000606	1.000082	1.001537	1.005875	1.013386		0.999936
17	1.011342	0.999536	1.022437	0.998885	0.999586	1.000106	1.017665
18	0.957152	1.007723	0.98365	0.841738	1.034538	0.165458	0.996892
19		1.000372	0.999651	0.999718	1.024941	0.999251	0.999285

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]
[1,]	0.9916861	0.9995629	NaN	0.9982184	0.9940708	NaN	NaN
[2,]	0.9965580	0.9920630	1.0001969	1.0085315	0.9968651	0.9903956	0.9953322
[3,]	0.9932875	1.0104728	0.9933692	1.0368808	0.9933710	0.9877144	0.9938652
[4,]	NaN	0.9894681	0.9852576	0.9945638	0.9951858	NaN	NaN
[5,]	1.0046526	0.9998150	1.0096932	1.0258746	1.0257174	0.9980676	0.9996733
[6,]	0.9997205	0.9997641	1.0000349	1.0226274	0.9999612	0.9998756	0.9999722
[7,]	1.0015997	0.9997689	1.0390477	1.0278184	1.0203159	0.9994296	1.0286119
[8,]	0.9999853	1.0071965	1.0007259	1.0308040	1.0679098	1.0011990	1.0541725
[9,]	0.9983209	1.0129513	1.0382785	1.0239120	1.0072209	1.0012624	0.9983209
[10,]	0.9997785	1.0033158	1.0107425	1.0251400	1.0107194	0.9992249	1.0025428
[11,]	1.0288777	1.0310884	1.0664484	1.0257539	1.0311068	0.9987190	1.0000925
[12,]	1.0115434	1.0244474	1.0657452	1.0141772	0.9995670	NaN	0.9995120
[13,]	0.9996607	1.0453517	1.0387938	1.0022994	1.0055227	1.0214596	0.9987484
[14,]	1.0011471	1.0234769	1.0144873	1.0065291	1.0170367	1.0556992	0.9992701
[15,]	NaN	0.9990917	1.0216118	1.0000634	0.9999222	1.0011141	0.9994149
[16,]	1.0003290	1.0001326	1.0018215	1.0059742	1.0134638	NaN	0.9999211
[17,]	1.0105791	0.9995363	1.0228031	0.9988483	0.9995800	1.0000436	1.0175848
[18,]	0.9566351	1.0077026	0.9836444	0.8416404	1.0344661	0.1654578	0.9966679
[19,]	NaN	1.0001639	1.0000817	0.9998021	1.0251115	0.9998381	0.9991786
[20,]	NaN	0.9998496	0.9918180	0.9996776	1.0037566	0.9998767	1.0029367

v1-v7 are for 7 species and 1-19 for years from 2003-2021